Cross Validated

Comparing nested GLMs via chi-squared and loglikelihood

Asked 10 years, 11 months ago Modified 10 years, 11 months ago Viewed 35k times

9

I want to compare two GLMs with binomial dependent variables. The results are:

```
m1 <- glm(symptoms ~ 1,         data=data2)
m2 <- glm(symptoms ~ phq_index, data=data2)
```

The model test gives the following results:

```
anova(m1, m2)
        no AIC    logLik  LR.stat df  Pr(>Chisq)
m1      1  4473.9 -2236.0
m2      9  4187.3 -2084.7  302.62  8   < 2.2e-16 ***
```

I am used to comparing these kinds of models using chi-squared values, a chi-squared difference, and a chi-squared difference test. Since all other models in the paper are compared this way, and since I'd like to report them in a table together: why exactly is this model test different from my other model tests in which I get chi-squared values and difference tests? Can I obtain chi-squared values from this test?

Results from other model comparisons (e.g., GLMER), look like this:

```
    #Df AIC     BIC     logLik  Chisq   Chi     DF diff Pr(>Chisq)
 m3  13  11288   11393   -5630.9 392.16
 m4  21  11212   11382   -5584.9 92.02   300.14  8       0.001
```

r generalized-linear-model chi-squared-test

Share Cite Edit Follow Flag

edited Sep 11, 2013 at 17:51
gung - Reinstate Monica
147k 89 402 711

asked Sep 10, 2013 at 13:52
Torvon
1,113 4 12 23

1 To what "paper" do you refer? – whuber ♦ Sep 11, 2013 at 17:26

1 Answer

Sorted by: Highest score (default)

The "chi-square value" you're looking for is the deviance (-2*(log likelihood), at least up to an additive constant that doesn't matter for the purposes of inference. R gives you the log-likelihood above (`logLik`)

13

and the likelihood ratio statistic (`LR.stat`): the LR stat is twice the difference in the log-likelihoods (2*(2236.0-2084.7)).

I'm a little puzzled by your `anova` results, since they don't seem to match the format that I get when I run `anova()` on two `glm()` fits: in particular, `stats:::anova.glm` (the `anova` method for `glm` objects) doesn't print the AIC ... are `m1` and `m2` really `glm` objects? (e.g. try `class("m1")`)

Can you give us a reproducible example? Here's what I get for a simple example, modified from `?glm` :

```
## Dobson (1990) Page 93: Randomized Controlled Trial :
d.AD <- data.frame(counts=c(18,17,15,20,10,20,25,13,12),
                   outcome=gl(3,1,9),
                   treatment=gl(3,3))
glm1 <- glm(counts ~ outcome + treatment, family = poisson, data=d.AD)
glm0 <- update(glm1, . ~ 1)
```

Model comparison gives the residual deviances (your "chi-squared value") and the differences between them ...

```
anova(glm0,glm1,test="Chisq")
## Analysis of Deviance Table
##
## Model 1: counts ~ 1
## Model 2: counts ~ outcome + treatment
##   Resid. Df Resid. Dev Df   Deviance Pr(>Chi)
## 1         8    10.5814
## 2         4     5.1291  4     5.4523    0.244
```

(if I left out the `test="Chisq"` , I would get all of the above but without the p-value)

I see from your cross-posting at http://article.gmane.org/gmane.comp.lang.r.general/299377 that you're actually using `ordinal::clm` , in which case we do get output that looks like what you have above (it's important to be precise) ... the results are not identical because the model is slightly different, and you are given the log-likelihoods (=-deviance/2) rather than the deviances, but the difference between the deviances ("LR.stat") is similar.

```
library(ordinal)
clm1 <- clm(ordered(counts) ~
   outcome + treatment, family = poisson, data=d.AD)
clm0 <- update(clm1, . ~ 1)
anova(clm0,clm1)
##       no.par      AIC  logLik LR.stat df Pr(>Chisq)
## clm0      7    50.777 -18.389
## clm1     11    52.839 -15.419  5.9389  4     0.2038
```

Share Cite Edit Follow Flag

edited Sep 11, 2013 at 19:03

answered Sep 11, 2013 at 17:20

Ben Bolker
44.7k 3 119 169

Ben, thank you for this very helpful comment! You are right, I confused CLM and GLM -- I am using GLMs now, and get the output you describe above. Residual Deviances in my case are M1=9693 (resid. DF=12339) and M2=10945 (resid. DF=12347) (Deviance=-1252) with DF_diff=-8 (p<0.001). Does that mean I can report these two values as chi-square values in a paper, or would that be incorrect? – Torvon Sep 12, 2013 at 11:47

1 You can report them as chi-square values, but it would generally be more informative to report them as deviances; there are lots of different statistics that are chi-square distributed – Ben Bolker Sep 12, 2013 at 13:18

Thank you again. Although deviances are compared, is this a "chi-square difference test"? Would you know of any example where such a model is reported in a paper? I'm used to reporting chi-square, df, AIC (rmsea etc.), chi-square-diff and df-diff, and p. Not sure what to report in this case above. Thank you! – Torvon Sep 23, 2013 at 14:28

2 If you are working in a GLM(M) context then "chi-square" and "deviance" are synonymous as far as I can tell (but as I pointed out "deviance" is more precise). – Ben Bolker Sep 23, 2013 at 14:31

Last question: usually in my analyses the DF are 8 smaller in the more constrained model. In this particular case above, using the code you suggested, "residual df" of the freely estimated model m1 is 8 *smaller* than the constrained model m2 (12339 vs. 12347), although the fit is much better for m1 (deviance: 9693 vs 10945). How can that be? Thank you! – Torvon Sep 25, 2013 at 14:06